

Anatomically Accurate Head Model for Traumatic Brain Injury Research

Guided Studies Research Paper

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1. ABSTRACT

Traumatic brain injury (TBI) remains a major health concern, and the influence of intracranial pressure (ICP) on brain response during impact is not well understood due to limitations in existing experimental head models. This study presents the development of a physical surrogate head model capable of simulating controlled ICP conditions. This system consists of a 3D-printed skull, hydrogel brain surrogate, cerebrospinal fluid (CSF) substitute, and an embedded vascular network. Pressure within the system was regulated through syringe-based inputs and measured using pressure gauges.

Blunt impact testing was performed using a drop tower under multiple configurations, including unpressurized controls and pressurized conditions. Pressure gauge measurements were recorded before and after impact to assess system performance. Initial trials showed significant pressure loss due to fluid leakage within the model. After modifications to improve sealing and system integrity, later trials demonstrated improved pressure retention. Additional testing also showed pressure transfer and equilibration between the vascular network and the CSF.

These results support the feasibility of using a pressure-controlled surrogate head model to study ICP-related effects and provide a basis for future investigation of pressure-dependent brain mechanics under impact loading.

2. INTRODUCTION

Traumatic brain injury (TBI) is a major global health concern, affecting over 50 million individuals worldwide each year [30]. It is the most frequent cause of death in adults less than 45 years old and the highest cause of long-term disability, often severely diminishing the quality of life of victims [2,30]. Despite its prevalence and devastating consequences, there is still no existing treatment that is capable of reversing the primary brain damage TBI causes. Instead, current clinical strategies are largely limited to preventing secondary injury and managing symptoms.

TBI most commonly results from blunt impacts to the head, such as from unintentional falls, sports collisions, and motor vehicle accidents. It is believed that the external dynamic loads that are applied to the head as a result of such incidents directly injure neurons and their axons by deforming the brain tissue [14]. The severity of deformation, commonly characterized by strain, strain rate, and stress distributions within brain tissue, varies based on several factors, including impact velocity, direction, and duration [14,22]. Understanding how external forces are transmitted through the skull and translated into intracranial deformation is therefore critical for predicting injury outcomes and developing effective mitigation strategies.

To investigate the biomechanics of TBI, both experimental and computational methods have been developed. For example, wearable devices can measure linear and rotational accelerations of the head during impact, providing valuable data on real-world injury scenarios [12]. Additionally, finite element (FE) models of the human head have been utilized to estimate internal stress and strain distributions within neural tissue based on measured head kinematic data [14]. Though useful and often highly anatomically accurate, FE models are still only predictive models that need to be validated against experimental data. Since it is not always either practical or feasible to test the effects of various impact scenarios on human subjects, physical surrogate head models have been used to perform blunt impact experiments [6,7]. However, while these approaches have advanced the study of TBI biomechanics, they often overlook the role of internal physiological conditions that may influence the brain's mechanical response to impact.

One such physiological factor is intracranial pressure (ICP), which plays a critical role in governing brain motion, vascular dynamics, and the mechanical response of brain tissue to external loading. Changes in

ICP can alter the mechanical environment within the skull, influencing how forces are transmitted through the cerebrospinal fluid (CSF) and brain tissue during impact. Despite its importance, ICP is typically not treated as an actively controlled variable in experiments. As a result, the influence of varying ICP levels on brain deformation during impact remains poorly understood.

A physiologically relevant mechanism for varying intracranial pressure (ICP) is the Valsalva maneuver (VM), which involves forced exhalation against a closed airway and commonly occurs during activities such as heavy lifting or straining [25,26]. In healthy individuals, baseline ICP typically ranges from 5 to 16 mmHg, but during VM, it can increase by approximately 10 to 20 mmHg, with peak values reaching 20 to 30 mmHg or higher [18,23,28]. Among the phases of the maneuver, Phase II is of particular interest because it produces a sustained rise in ICP (approximately 15 mmHg above baseline), accompanied by reduced venous return and altered cerebral hemodynamics affecting mean arterial pressure (MAP) and cerebral perfusion pressure (CPP) [26]. For this reason, Phase II was selected as the primary condition for replication in this study. Elevated ICP under these conditions may alter brain motion within the skull during impact, potentially reducing dynamic loading on brain tissue and motivating investigation of possible pressure-mediated neuroprotective effects.

Despite this potential, existing surrogate head models rarely incorporate a system to manipulate internal pressure levels, making studying ICP dynamics in a controlled environment not feasible. Most models are simplified physical systems that do not fully capture the interactions between brain tissue, vasculature, and pressure. Particularly, there is a lack of integrated vasculature in existing physical models that limits the ability to study the interactions between fluid and brain tissue that are critical to understanding ICP-related effects.

To address these limitations, this work focuses on the development of an anatomically accurate physical head model that incorporates representations of the skull, dura mater, scalp, cerebrospinal fluid (CSF), brain tissue, and vasculature. The model is equipped with pressure sensors and syringes to allow controlled pressure input and measurement to serve as a platform for investigating intracranial pressure dynamics under Valsalva-like conditions. By balancing anatomical realism with fabrication feasibility, this system aims to provide a practical approach for studying pressure-related brain behavior in a controlled setting.

3. METHODS

3.1 Skull Design & Fabrication

A CT scan-derived skull model obtained from NIH3D was modified in Fusion360 and scaled to match the head breadth and length of a 50th percentile U.S. male based on published anthropometric data [29]. The model was 3D printed in PA12 nylon using an HP Multi Jet Fusion printer. Nylon was selected because it has a tensile strength of approximately 48 MPa and a tensile modulus of 1700–1800 MPa, which fall within the reported range for human cranial bone (tensile strength: 24.82–62.05 MPa; tensile modulus: 1230–5380 MPa) [6, 7].

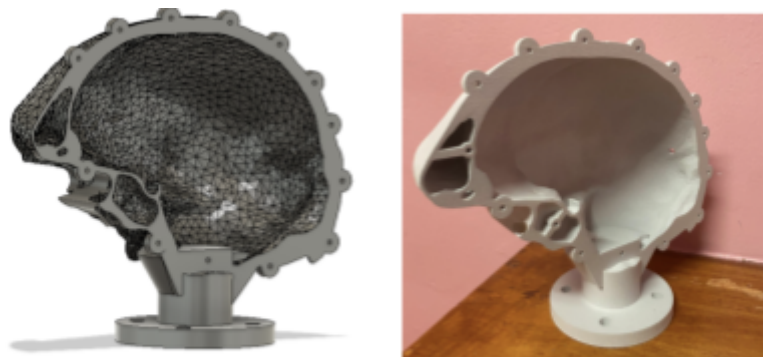


Figure 1. CAD skull model (left). Nylon 3D printed skull (right).

The skull geometry was initially developed using two configurations to support both vasculature integration and comparative testing. A 3/4 skull model was designed to accommodate a 3D-printed Circle of Willis and provide sufficient internal volume for integrating complex vascular structures. In parallel, a simplified 1/2 skull model without vasculature was considered as a control to evaluate whether the inclusion of vasculature would significantly impact system behavior.

However, due to challenges associated with fabricating and integrating the 3D printed vascular network, the vasculature approach was modified to utilize tubing rather than a fully printed structure. As a result, the design was consolidated to a 1/2 skull model to maintain consistency. This geometry maintains the potential for future comparisons between models with and without vasculature.

3.2 Brain Fabrication

Among most existing models, polyacrylamide (PAA) hydrogels have been identified as the most anatomically accurate surrogate for human brain tissue due to their tunable viscoelastic properties [5,10]. Literature values indicate that PAA gels with concentrations of approximately 10wt% and 12wt% can mimic the anatomical behavior of gray and white matter, respectively, with reported storage modulus values of ~1400 Pa for gray matter and ~2000 Pa for white matter which ultimately closely matches those of human brain tissue [5,10].

Thus, the brain model was fabricated using polyacrylamide (PAA) hydrogel cast within a custom CAD designed mold as shown in Figure 2. A three-quarter (3/4) brain geometry was initially developed in *Fusion 360*

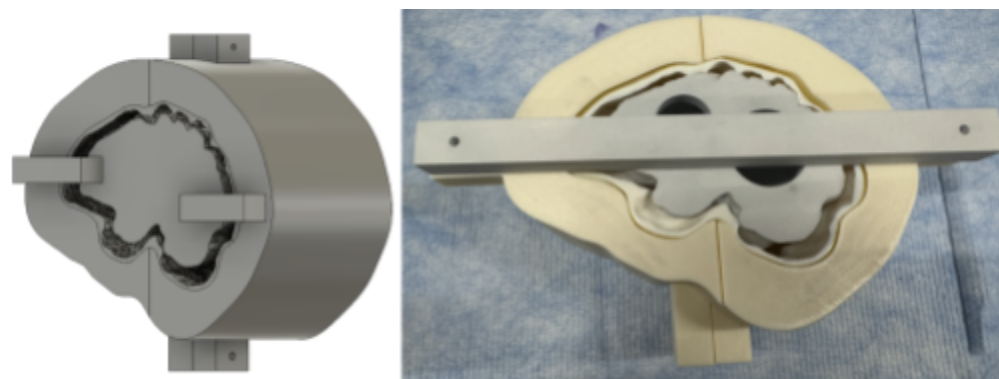


Figure 2. Brain mold 3D printed (right) based on CAD model (left) consisting of a flexible mold, a rigid body for structure, and a white matter core.

based on an MRI scan of a human male brain. The model was scaled to fit within the skull with a ~2 mm gap for the cerebral spinal fluid (CSF) and adjustments were also made to maintain realistic proportions of gray and white matter. Specifically, the white matter mold was scaled to ~85% of the overall brain geometry to achieve a white matter volume of ~60% of the total brain volume, which is consistent with reported anatomical data [16].

The mold consists of a flexible outer shell, a white matter core, and a rigid support structure as depicted in Figure 2. The outer mold was 3D printed using a thin flexible elastic material (Agilus30), allowing for ease of demolding while preserving fine geometric features. A rigid external structure was used to maintain the structural integrity of the flexible mold and ensure dimensional stability during casting. The white matter core insert was fabricated from a rigid, non-porous material to prevent

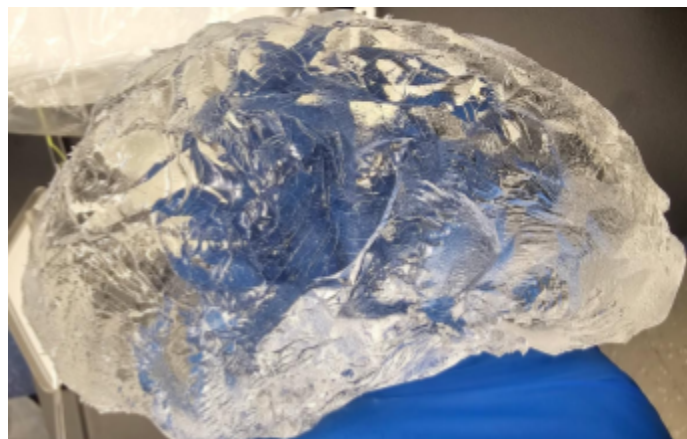


Figure 3. Fabricated hydrogel brain surrogate tuned for gray and white matter.

fluid absorption and maintain clean separation during casting.

To facilitate demolding and prevent the hydrogel from sticking, all mold components were thoroughly coated with sodium dodecyl sulfate (SDS) prior to casting.

The hydrogel brain was fabricated in two stages to distinguish gray and white matter as depicted in Figure 3. First, the gray matter layer was cast using a 10 wt% PAA hydrogel solution. Once fully cured, the inner white mold core insert was removed, and a 12 wt% PAA hydrogel solution was poured to form the white matter region. The hydrogel was prepared using deionized water, acrylamide powder, N,N'-methylenebisacrylamide (MBA) as a crosslinker, ammonium persulfate (APS) as an initiator, and TEMED as an accelerator, with ratios selected based on literature-recommended values to approximate brain tissue mechanics [5,10]. Detailed hydrogel formulations are provided in Table 1.

Table 1. Polyacrylamide (PAA) hydrogel formulations used to fabricate gray and white matter regions.

Component	Function	White Matter (12 wt%)	Gray Matter (10 wt%)
Deionized Water	Solvent	525 mL	350 mL
Acrylamide	Monomer	63 g	35 g
MBA (N,N' – methylenebisacrylamide)	Crosslinker	1.05 g	0.583 g
APS (Ammonium Persulfate)	Initiator	0.525 g	0.35 g
TEMED	Accelerator	0.525 mL	0.35 mL
Titanium Dioxide (TiO ₂)	Opacifier	1.5 g	1 g

In order to improve the visualization of the hydrogel structure, multiple dyeing approaches were explored. Ultimately, titanium dioxide (TiO₂) powder was incorporated into the hydrogel formulation which produced a uniformly opaque material while maintaining successful curing. This opacity significantly improved visual identification for experimental markers. A stencil was then used to apply a grid of tracking points onto the brain surface using a black marker, with points spaced 5 mm apart to enable high-speed motion tracking as seen in Figure 4.

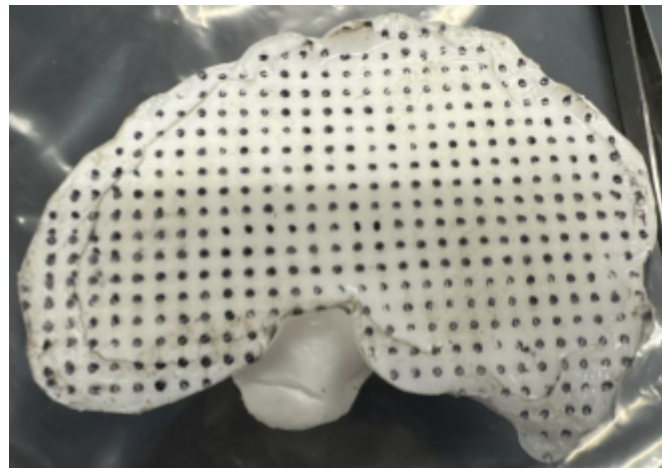


Figure 4. Opaque brain without vasculature with grid markings for high speed tracking.

3.3 Vasculature Design

Existing physical models lack integrated vasculature which in turn fails to capture the complex fluid to structure interactions associated with ICP changes. Approaches for modeling vasculature were also investigated. One method in particular involves the use of a dissolvable sacrificial 3D printed core which would be embedded in the soft brain material which would create hollow channels that

represent the complex vascular networks [19]. However, implementation of this approach was not feasible at the time of the study due to limitations of the available equipment.

The vasculature design underwent multiple iterations to balance anatomical accuracy, material behavior, and fabrication feasibility. Two primary approaches were explored: a 3D printed Circle of Willis (CoW) model and a simplified tubing-based system.

3.3.1 Circle of Willis Modeling

An anatomical model of the Circle of Willis was developed based on CTA scan data from University of Michigan [1]. Vessel diameters were designed to fall within reported physiological ranges, with inner diameters ranging from approximately 2–4 mm depending on the vessel type [15,17]. The model included major arterial branches such as the basilar artery, internal carotid arteries, and anterior, middle, and posterior cerebral arteries.

Initial fabrication efforts focused on 3D printing the vasculature using flexible materials to replicate vessel compliance. Figure 4 displays a 3D printing trial of the Circle of Willis. Two primary materials were evaluated: Agilus30 and Elastico. While Agilus30 more closely matched the mechanical softness of biological tissue, it presented significant challenges during post-processing. Specifically, the material was difficult to clean, and thin vascular features were prone to tearing during support removal and handling.



Figure 4. 3D printed Circle of Willis.

Elastico was subsequently tested as an alternative due to its higher stiffness and improved structural integrity. This material allowed for more reliable printing and reduced tearing; however, its increased rigidity made integration into the hydrogel brain model more difficult. Additionally, neither material provided an ideal balance between flexibility, durability, and ease of incorporation into the casting process.

Due to these limitations, direct implementation of a fully 3D printed vascular network within the hydrogel model was determined to be impractical within the current fabrication constraints.

3.3.2 Transition to Tubing

Given the challenges associated with 3D printed vasculature, the design was adapted to utilize commercially available silicone tubing as a simplified representation of cerebral vessels. Tubing was selected based on inner diameter, wall thickness, and mechanical properties to approximate physiological conditions while maintaining manufacturability.

The selected tubing had an inner diameter of approximately 0.3 mm and a Young's modulus on the order of 2.5–3.5 MPa, which falls within the upper range of reported cerebral artery stiffness [4]. While this does not fully replicate the compliance of real vasculature, it provides a functional system capable of supporting fluid flow and pressure transmission.

To evaluate whether the selected tubing configuration could support physiologically relevant pressure conditions, a simplified flow analysis was performed using the Hagen–Poiseuille equation for laminar flow:

$$L = \frac{\Delta P \pi r^4}{8 \mu Q}$$

Where L is the tube length, ΔP is the pressure difference, r is the inner radius of the tubing, μ is the dynamic viscosity of the fluid, and Q is the volumetric flow rate. Water was assumed as the working fluid with a viscosity of $\mu = 0.001 Pa \cdot s$ and pressure increases were set in the range of 17-30 mmHg as this is what is experienced during the Valsalva maneuver [18,23,28].

Using this model, the maximum allowable tube length was calculated across a range of flow rates to determine whether a 1m tube length would be feasible within the expected pressure range. These results are summarized in Table 2.

Table 2. Maximum allowable tube length as a function of flow rate.

Flow Rate (mL/min)	$L_{\max}$ @ 17 mmHg (m)	$L_{\max}$ @ 30 mmHg (m)
1.0	1.79	1.96
0.5	3.58	3.92
0.1	17.91	19.61
0.05	35.83	39.23
0.01	179.13	196.13

The results indicate that at lower flow rates, significantly longer tubing lengths can be supported without exceeding the target pressure range, while higher flow rates impose limitations in length. For example, at a flow rate of 1 mL/min, the maximum allowable tube length is approximately 1.8–2.0 m, whereas at lower flow rates, allowable lengths increase substantially. Based on these results, a tube length of 1 m was selected, as it remains within the feasible range across all considered conditions.

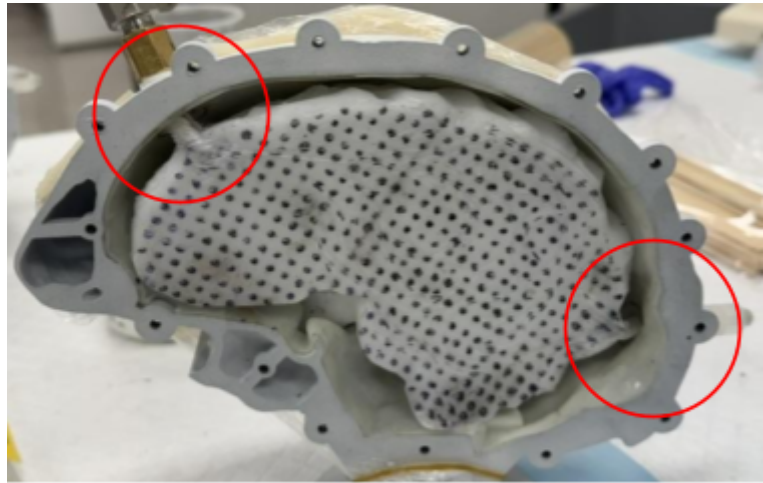


Figure 6. Brain surrogate with integrated vascular tubing network consisting of 40 silicone tubes (1 m length each), distributed between gray matter ($n = 30$) and white matter ($n = 10$), and connected through a luer interface.

In addition to flow feasibility, geometric constraints were also considered when integrating tubing into the system. A luer connection with a diameter of approximately 6 mm was used as the interface, and the number of tubes that could be accommodated was estimated based on cross-sectional area calculations. While the theoretical maximum number of tubes was approximately 89, practical considerations such as spacing and sealing reduced the usable number to approximately 40 as depicted in Figure 6.

To better reflect anatomical trends, the distribution of tubing was scaled between gray and white matter regions based on reported vascular density differences, with gray matter containing approximately 3–4 times more vasculature than white matter [16]. This resulted in an estimated allocation of approximately 30 tubes in gray matter and 10 tubes in white matter.

3.4 Dura Mater Fabrication

A 3D model of the dura mater was created using the inner surface of the final edited version of the skull in ZBrush. It was then 3D printed in Elastico using a Stratasys PolyJet printer and was 1.10 mm thick, which is in the range of reported human dura mater thickness (1.05 ± 0.22 mm) [3]. Elastico was selected because it has a tensile strength of approximately 435–725 psi and a density of 1.1–1.3 g/cm³, which fall within the reported range for human dura mater (tensile strength: 435–1595 psi; density: 1.05–1.55 g/cm³) [8, 31]. The 3D-printed dura mater was glued onto the inner surface of the nylon skull as depicted in Figure 7.

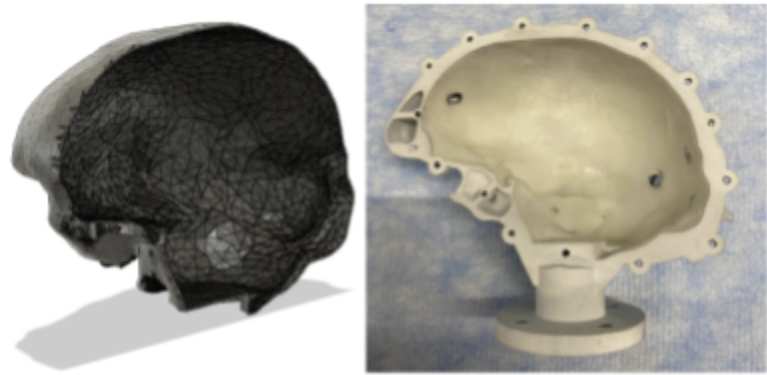


Figure 7. CAD model dura mater (left). 3D printed dura mater (right).

3.5 Scalp Fabrication

Smooth-On Vytaflex 40 was used as the scalp simulant due to its similar mechanical properties to human scalp. Vytaflex 40 has a density of approximately 1000–1030 kg/m³, which is comparable to that of human scalp tissue (~1000 kg/m³), and a Shore A hardness of 40, which falls within the reported range for human skin (Shore A 20–75). Additionally, published research has shown that 2 mm or 4 mm thick pads of Vytaflex 40 or Vytaflex 50 are the most biofidelic in simulating the human scalp [11].

A box was laser cut out of an acrylic sheet and used to mold Vytaflex 40 into 2 mm thick pads. The pads were then epoxied onto the outer skull only on the sides and corners as seen in Figure 8. The center of the scalp was not attached to the skull in order to allow for a sliding motion similar to that of a real human scalp.

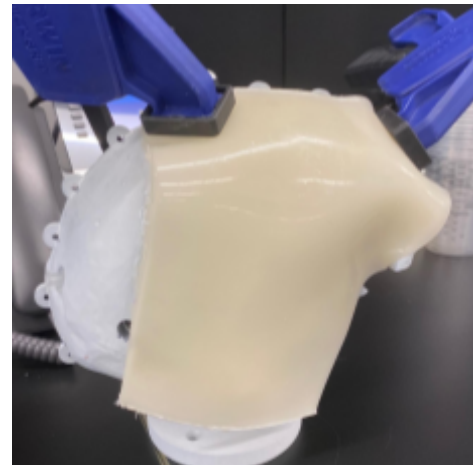


Figure 8. Placement of scalp on the skull.

3.6 CSF Modeling

Cerebrospinal fluid (CSF) was modeled using deionized water to replicate fluid behavior within the cranial cavity. Water was selected as it has similar viscosity to human CSF and due to its availability, ease of use, and ability to approximate the incompressible nature of CSF for pressure-based experiments [21,24].

3.7 System Integration

The final system was assembled by integrating the skull, dura mater, hydrogel brain, vasculature network, scalp, and cerebrospinal fluid (CSF) components into a single platform. The dura mater layer was affixed to the inner surface of the skull, while the scalp simulant was attached externally. The hydrogel brain was positioned within the cranial cavity with an approximate 2 mm gap, which was filled with water to represent the CSF.

In the vascularized configuration, the skull was designed with two pressure inlet ports, each controlled via a syringe: (1) one dedicated to the vasculature network and (2) the other to the CSF compartment. Two additional ports were incorporated for pressure gauge attachment, enabling independent measurement of vascular and CSF pressures. The embedded silicone tubing network was routed through a luer connection to facilitate fluid input and pressure regulation.

For the configuration without vasculature, the system was simplified to include a single pressure inlet port connected to the CSF compartment, along with one additional port for pressure gauge attachment to monitor intracranial pressure. This configuration enabled isolated control and measurement of CSF pressure.

The assembled system configurations are shown in Figure 9.

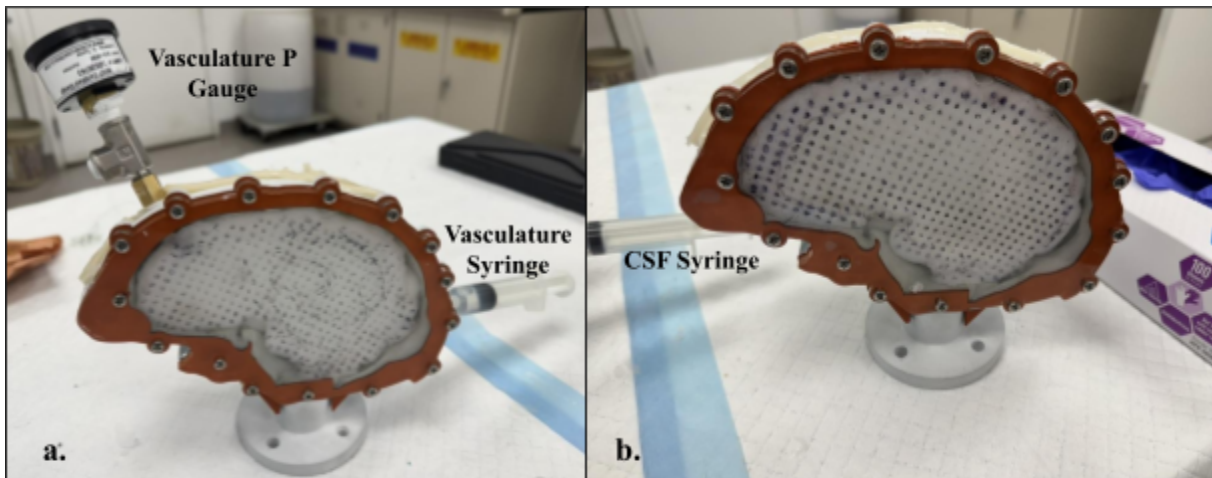


Figure 9. Surrogate head model configurations showing (a) system with integrated vasculature and CSF, equipped for independent regulation of vascular and intracranial (CSF) pressures, and (b) system without vasculature, configured for CSF pressure control only.

3.8 Blunt Impact Testing

Blunt impact testing was conducted using a drop tower apparatus to evaluate the response of the surrogate head model under controlled impact conditions. The head model was mounted onto a base plate positioned beneath the drop mechanism. Four experimental configurations were tested:

1. An unpressurized control model without vasculature
2. A model with CSF pressure applied (no vasculature)
3. An unpressurized control model with embedded vasculature
4. A model with applied vascular pressure (with embedded vasculature)

For each test, the desired pressure was applied using a syringe through designated inlet ports. In the vascularized configuration, CSF and vascular pressures were independently controlled and monitored

using separate pressure gauges. In the non-vascular configuration, only CSF pressure was applied and measured.

Baseline pressures were recorded prior to impact, and pressure readings were recorded again immediately after impact to quantify pressure changes and assess system integrity, including potential leakage. Multiple trials were conducted across a range of initial pressures for each configuration. High-speed imaging was used to capture brain motion during impact.

4. RESULTS

4.1 Pressure Gauge Readings

Pressure measurements before and after impact for both the CSF-only and vascularized configurations are summarized in Table 3.

In the CSF-only configuration (Position 1), pressure drops were consistently observed following impact, with decreases ranging from approximately 0.6 to 1.0 psi. One trial resulted in complete pressure loss due to leakage, indicating limitations in system sealing during early testing.

In the vascularized configuration (Position 2), initial trials exhibited significant pressure loss, including complete drops to 0 psi, which were attributed to leakage from the syringe interface and base connections. After tightening system components and improving sealing, later trials demonstrated improved pressure retention. In these trials, pressure drops were reduced to approximately 0.1–0.3 psi, with one trial showing no observable leakage.

The effect of system adjustments is evident when comparing early and later trials, indicating improved structural integrity and pressure stability following repositioning of the base plate and reinforcement of connections.

Table 3. Pressure measurements before and after impact for CSF-only and vascularized configurations.

Position	Configuration	Trial	Before (psi)	After (psi)	Notes
1	CSF only	1	1.7	1.0	—
1	CSF only	2	2.5	1.9	—
1	CSF only	—	1.9	0.0	Leak
1	CSF only	3	2.0	1.4	—
2	Vascularized	1	1.7	0.0	—
2	Vascularized	2	1.7	0.0	Syringe leak, bottom leak
2	Vascularized	3	1.9	1.6	Minor leak
2	Vascularized	4	3.1	2.7	No leak
2	Vascularized	—	2.4	2.2	Did not trigger
2	Vascularized	—	2.2	2.0	Did not trigger
2	Vascularized	5	1.9	1.8	—

2	Vascularized	6	1.6	1.4	—
2	Vascularized	7	2.0	1.9	—

Simultaneous vascular and CSF pressure measurements for the vascularized configuration are presented in Table 4. These results show that the system approaches pressure equilibrium, as evidenced by similar pressure values between the vascular and CSF compartments across trials.

Table 4. Simultaneous vascular and CSF pressure measurements demonstrating system equilibration.

Position	Configuration	Trial	Vascular (psi)	CSF (psi)	Notes
2	Vascularized	1	1.4	1.4	System equilibrating
2	Vascularized	2	1.7	1.7	
2	Vascularized	3	2.0	1.9	

4.2 High Speed Image Tracking (Preliminary Implementation)

High-speed imaging was implemented to capture brain motion during impact; however, quantitative analysis of deformation was not completed within the scope of this study. A grid of surface markers was successfully applied to the brain surrogate to enable future motion tracking and strain analysis.

5. DISCUSSION

5.1 Design Tradeoffs

The development of the surrogate head model required balancing anatomical accuracy with fabrication feasibility and experimental functionality. While the initial design aimed to incorporate a fully 3D-printed Circle of Willis, limitations in available materials and fabrication methods made it difficult to produce structures that were both mechanically compliant and structurally robust. As a result, the design was adapted to utilize silicone tubing to represent vasculature. This approach sacrificed anatomical fidelity in terms of geometry and branching complexity but provided a practical and reliable method for enabling fluid flow and pressure control within the system. Additionally, the implementation of luer-based pressure interfaces enabled controllable and measurable pressure input but introduced potential sources of leakage and mechanical instability.

Overall, these tradeoffs reflect a design approach focused on achieving a functional experimental platform while maintaining a reasonable level of anatomical relevance.

5.2 Limitations

Several limitations were identified during the development and testing of the surrogate head model. The most significant limitation was system leakage, particularly in early trials, which resulted in substantial pressure loss and, in some cases, complete failure to maintain pressure. Leakage was observed at syringe interfaces, tubing connections, and base plate junctions, indicating that sealing and structural integrity are critical factors in system performance.

The simplified representation of vasculature using straight silicone tubing does not fully capture the complex geometry, compliance, and branching behavior of real cerebral vessels. Similarly, the CSF was modeled as static deionized water, which does not account for dynamic fluid flow or physiological regulation present in vivo.

6. FUTURE WORK

Future work will focus on improving system reliability, expanding experimental capabilities, and increasing anatomical fidelity. Enhancements to sealing methods and connection interfaces will be implemented to minimize leakage and enable consistent pressure control across trials.

Further development of the vasculature model may include more anatomically accurate geometries, such as branched networks or hybrid approaches combining tubing with molded channels. Additionally, integrating dynamic CSF flow and more realistic fluid–structure interactions will improve physiological relevance.

Future experiments will incorporate high-speed imaging and quantitative tracking of brain deformation using the applied surface marker grid. This will enable analysis of strain, displacement, and motion patterns under varying pressure conditions.

Ultimately, this platform can be used to investigate the role of intracranial pressure in modulating brain injury and to explore potential pressure-mediated protective mechanisms under blunt impact conditions.

7. CONCLUSION

This study presents the design and development of an anatomically accurate surrogate head model capable of simulating controlled intracranial pressure conditions. The system integrates key components, including a hydrogel brain surrogate, CSF substitute, and an embedded vascular network, with the ability to independently regulate and measure pressure.

Preliminary testing demonstrated the feasibility of applying and monitoring pressure within the system, while also highlighting challenges related to leakage and structural integrity. Iterative improvements to the setup resulted in enhanced pressure retention and system stability.

Overall, this work establishes a functional experimental platform for investigating pressure-dependent brain behavior under blunt impact. The findings highlight both the potential and the challenges of incorporating controlled pressure conditions into physical head models and also provide a foundation for future studies aimed at understanding the role of intracranial pressure in traumatic brain injury.

REFERENCES

- [1] Alberto, Figueroa C. “.Stl File of Circle of Willis.” Deep Blue Data, 2020, deepblue.lib.umich.edu/data/concern/data_sets/nv935310f, <https://doi.org/10.7302/xx1r-zg70>.
- [2] Coronado VG, Xu L, Basavaraju SV, et al. Surveillance for traumatic brain injury-related deaths--United States, 1997-2007. *MMWR Surveill Summ*. 2011;60(5):1-32.
- [3] De Kegel D, Vastmans J, Fehervary H, Depreitere B, Vander Sloten J, Famaey N. Biomechanical characterization of human dura mater. *J Mech Behav Biomed Mater*. 2018;79:122-134. doi:10.1016/j.jmbbm.2017.12.023
- [4] Ebrahimi AP. Mechanical properties of normal and diseased cerebrovascular system. *J Vasc Interv Neurol*. 2009 Apr;2(2):155-62. PMID: 22518247; PMCID: PMC3317338.
- [5] Galindo EJ, et al. Simultaneous High-Frame-Rate Acoustic Plane-Wave and Optical Imaging of Intracranial Cavitation in Polyacrylamide Brain Phantoms during Blunt Force Impact. *Bioengineering (Basel)*. 2024;11(2).
- [6] Hanna M, et al. Anatomical Features and Material Properties of Human Surrogate Head Models Affect Spatial and Temporal Brain Motion under Blunt Impact. *Bioengineering (Basel)*. 2024;11(7).
- [7] Hanna M, et al. A Method for Evaluating Brain Deformation Under Sagittal Blunt Impacts Using a Half-Skull Human-Scale Surrogate. *J Biomech Eng*. 2023;145(6).
- [8] Harmukh A, Ganpule SG. Evaluation of Snowboarding Helmets in Mitigation of the Biomechanical Responses of Head Surrogate. *Applied Sciences*. 2024;14(23):11460. <https://doi.org/10.3390/app142311460>
- [9] Kerwin J, Yücesoy A, Vidhate S, Dávila-Montero BM, Van Orman JL, Pence TJ, Tartis M, Mejía-Alvarez R, Willis AM. Sulcal Cavitation in Linear Head Acceleration: Possible Correlation With Chronic Traumatic Encephalopathy. *Front Neurol*. 2022;13:832370. doi:10.3389/fneur.2022.832370
- [10] Knutsen AK, Vidhate S, McIlvain G, et al. Characterization of material properties and deformation in the ANGUS phantom during mild head impacts using MRI. *J Mech Behav Biomed Mater*. 2023;138:105586. doi:10.1016/j.jmbbm.2022.105586
- [11] Li Y, Vakiel P, Adanty K, et al. Influence of surrogate scalp material and thickness on head impact responses: Toward a biofidelic head-brain physical model. *J Mech Behav Biomed Mater*. 2023;142:105859. doi:<https://doi.org/10.1016/j.jmbbm.2023.105859>
- [12] Liu Y, Domel AG, Cecchi NJ, et al. Time Window of Head Impact Kinematics Measurement for Calculation of Brain Strain and Strain Rate in American Football. *Ann Biomed Eng*. 2021;49(10):2791-2804. doi:10.1007/s10439-021-02821-z
- [13] Mandell DM, Mossa-Basha M, Qiao Y, et al. Intracranial Vessel Wall MRI: Principles and Expert Consensus Recommendations of the American Society of Neuroradiology. *AJNR Am J Neuroradiol*. 2017;38(2):218-229. doi:10.3174/ajnr.A4893
- [14] Meaney DF, Morrison B, Dale Bass C. The mechanics of traumatic brain injury: a review of what we know and what we need to know for reducing its societal burden. *J Biomech Eng*. 2014;136(2):021008. doi:10.1115/1.4026364

- [15] Mirza M, et al. Variability in Intracranial Vessel Diameters and Considerations for Neurovascular Models: A Systematic Review and Meta-Analysis. *Stroke: Vascular and Interventional Neurology*. 2024;4(4):e001177.
- [16] Mitsuhashi M, Van Epps D, Sun H, et al. White Matter Matters: New Approach to the Brain's Hidden Half Using Circulating Oligodendrocyte-Derived Extracellular Vesicles. *Cells*. 2025;14(22):1771. doi:10.3390/cells14221771
- [17] Mujagić S. The Inner Diameter of Arteries of the Circle of Willis Regarding Gender and Age on Magnetic Resonance Angiography. *Acta Medica Saliniana*. 2013;42:6-12.
- [18] Neville L, Egan RA. Frequency and amplitude of elevation of cerebrospinal fluid resting pressure by the Valsalva maneuver. *Can J Ophthalmol*. 2005;40(6):775-777.
- [19] Nilsson DPG, et al. Patient-specific brain arteries molded as a flexible phantom model using 3D printed water-soluble resin. *Sci Rep*. 2022;12(1):10172.
- [20] Pennacchietti V, et al. Maneuver protocol for outpatient telemetric intracranial pressure monitoring in hydrocephalus patients. *Childs Nerv Syst*. 2023;39(1):185-195.
- [21] Petrone N, Candiottio G, Marzella E, et al. Feasibility of using a novel instrumented human head surrogate to measure helmet, head and brain kinematics and intracranial pressure during multidirectional impact tests. *J Sci Med Sport*. 2019;22 Suppl 1:S78-S84. doi:10.1016/j.jsams.2019.05.015
- [22] Post A, Hoshizaki TB, Gilchrist MD, et al. Traumatic brain injuries: the influence of the direction of impact. *Neurosurgery*. 2015;76(1):81-91. doi:10.1227/NEU.0000000000000554
- [23] Prabhakar H, et al. Intracranial pressure changes during Valsalva manoeuvre in patients undergoing a neuroendoscopic procedure. *Minim Invasive Neurosurg*. 2007;50(2):98-101.
- [24] Roldan M, Kyriacou PA. Head Phantom for the Acquisition of Pulsatile Optical Signals for Traumatic Brain Injury Monitoring. *Photonics*. 2023;10(5).
- [25] Themes UF. Chapter 41 – Valsalva Manoeuvre. *Anesthesia Key*. 27 Sept. 2020.
- [26] Tiecks FP, et al. Effects of the Valsalva Maneuver on Cerebral Circulation in Healthy Adults. *Stroke*. 1995;26(8):1386-1392.
- [27] Wermer A, et al. Materials Characterization of Cranial Simulants for Blast-Induced Traumatic Brain Injury. *Military Medicine*. 2020;185(Supplement_1):205-213.
- [28] Yiangou A, et al. The Impact of Valsalva Manoeuvres and Exercise on Intracranial Pressure and Cerebrovascular Dynamics in Idiopathic Intracranial Hypertension. *Neuroophthalmology*. 2024;48(2):122-133.
- [29] Young JW. *Head and Face Anthropometry of Adult U.S. Civilians*. Federal Aviation Administration Washington DC Office of Aviation Medicine; 1993.
- [30] Zhong H, Feng Y, Shen J, et al. Global Burden of Traumatic Brain Injury in 204 Countries and Territories From 1990 to 2021. *Am J Prev Med*. 2025;68(4):754-763. doi:10.1016/j.amepre.2025.01.001

- [31] Zwirner J, Scholze M, Waddell JN, Ondruschka B, Hammer N. Mechanical Properties of Human Dura Mater in Tension - An Analysis at an Age Range of 2 to 94 Years. *Sci Rep.* 2019;9(1):16655. doi:10.1038/s41598-019-52836-9